GROUP WORK PROJECT # 1
GROUP NUMBER: 16855

MScFE 622: Stochastic Modeling

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Executive summary

This report presents the assumptions, results, interpretations, and figures for all project questions. SM Energy spot is $232.90; the annual risk-free rate is 1.50%; one year is 250 trading days; and no dividend yield is assumed.

Step 1 - 15-day derivative

1(i) Heston calibration using Lewis (2001)

Heston represents equity returns with a randomly changing, mean-reverting variance process that can reproduce volatility clustering and implied-volatility skew. Lewis converts its characteristic function into a one-dimensional Fourier pricing integral.

Parameters (Equations 1-3). S_t is the stock price and v_t is its instantaneous variance at time t; r is the continuously compounded risk-free rate and dt is an infinitesimal time increment. W_t^S and W_t^v are Brownian motions driving price and variance. The variance parameters are mean-reversion speed kappa, long-run variance theta, volatility of variance sigma, correlation rho between the Brownian shocks, and initial variance v_0.

$$dS_t = rS_t\,dt + \sqrt{v_t}S_t\,dW_t^S. \tag{1}$$

$$dv_t = \kappa(\theta - v_t)\,dt + \sigma\sqrt{v_t}\,dW_t^v. \tag{2}$$

$$dW_t^S\,dW_t^v = \rho\,dt. \tag{3}$$

Parameters (Equation 4). phi_T(z) is the characteristic function of the log return over maturity T at complex Fourier argument z; i=sqrt(-1), S_0 and S_T are the initial and terminal stock prices, and E^Q denotes expectation under the risk-neutral measure Q.

$$\varphi_T(z) = \mathbb{E}^{\mathbb{Q}}[e^{iz\log(S_T/S_0)}]. \tag{4}$$

Parameters (Equations 5-6). C_0 and P_0 are today's European call and put values; S_0 is spot, K is strike, T is time to maturity, and r is the risk-free rate. u is the Fourier integration variable, i=sqrt(-1), Re[dot] takes the real part, pi is the circle constant, and phi_T is the Heston characteristic function from Equation (4).

$$C_0 = S_0 - \frac{e^{-rT}\sqrt{S_0K}}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{iu\log(S_0/K)}\varphi_T(u-i/2)}{u^2+1/4}\right]du. \tag{5}$$

$$P_0 = C_0 - S_0 + Ke^{-rT}. \tag{6}$$

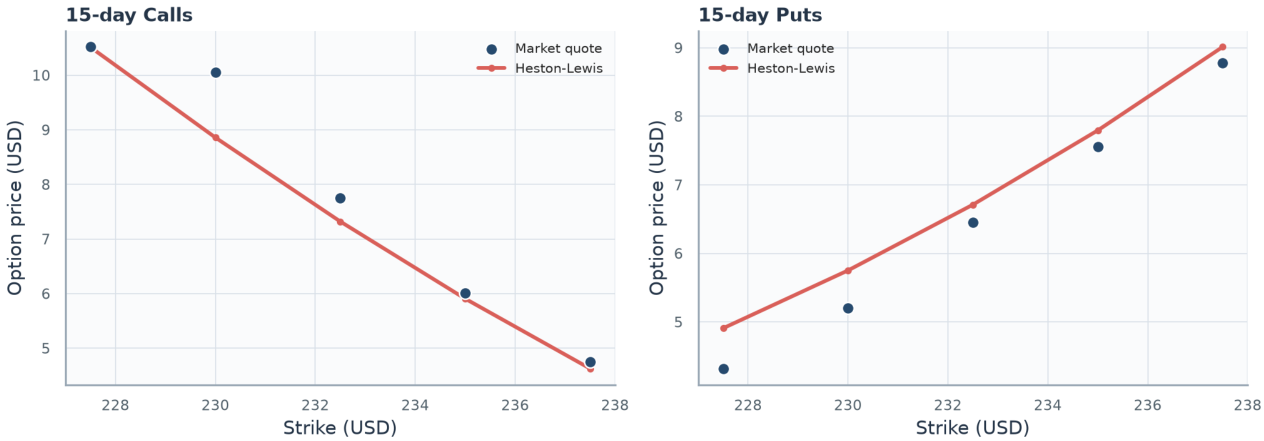
Parameters (Equation 7). Theta is the trial Heston parameter vector; N=10 is the number of option quotes and i indexes them. V_i^model(Theta) and V_i^market are the model and observed USD prices, respectively, and MSE is their mean squared pricing error.

$$\mathrm{MSE}(\Theta) = \tfrac{1}{N}\sum_{i=1}^{N}\left(V_i^{\mathrm{model}}(\Theta) - V_i^{\mathrm{market}}\right)^2, \qquad N = 10. \tag{7}$$

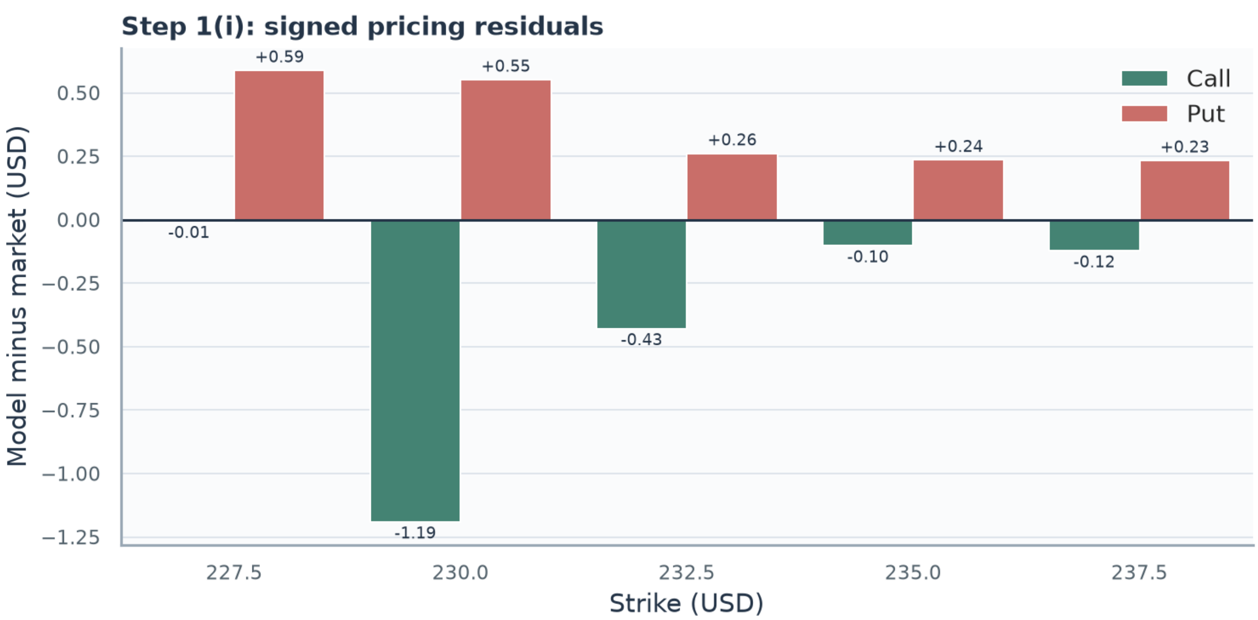
The 15-day calls and puts were fitted jointly using the Heston specification and Lewis representation (Heston 327-343; Lewis), with unweighted USD price MSE as required by the project brief (WorldQuant University 1-2). The fitted parameters are kappa=2.2032, theta=0.0010, sigma=1.2733, rho=-0.9444, and initial variance=0.1084. MSE is 0.2458 and RMSE is $0.4958.

The input quotes have a put-call parity RMSE of $0.9068; an arbitrage-consistent fit cannot make all residuals zero. Short maturity also weakens identification of long-run variance and mean reversion.

Initial variance implies 32.93% annualized volatility and rho=-0.9444 indicates a strong leverage effect. Theta is at its lower bound and the Feller diagnostic is -1.6170; the parameter vector should therefore be treated as a short-horizon fit rather than a long-run forecast.



Plot 1. 15-day market and model prices. The fitted curves follow the market points closely for both calls and puts, while the remaining gaps reflect quote inconsistency and calibration error.



Plot 2. 15-day residuals. Bars above zero indicate model overpricing and bars below zero indicate underpricing; the mix of signs shows that no single directional bias explains the errors.

1(ii) Carr-Madan comparison

Carr-Madan damps the option-price function so it can be evaluated through an integrable Fourier transform, providing an independent pricing and calibration route for the same Heston dynamics.

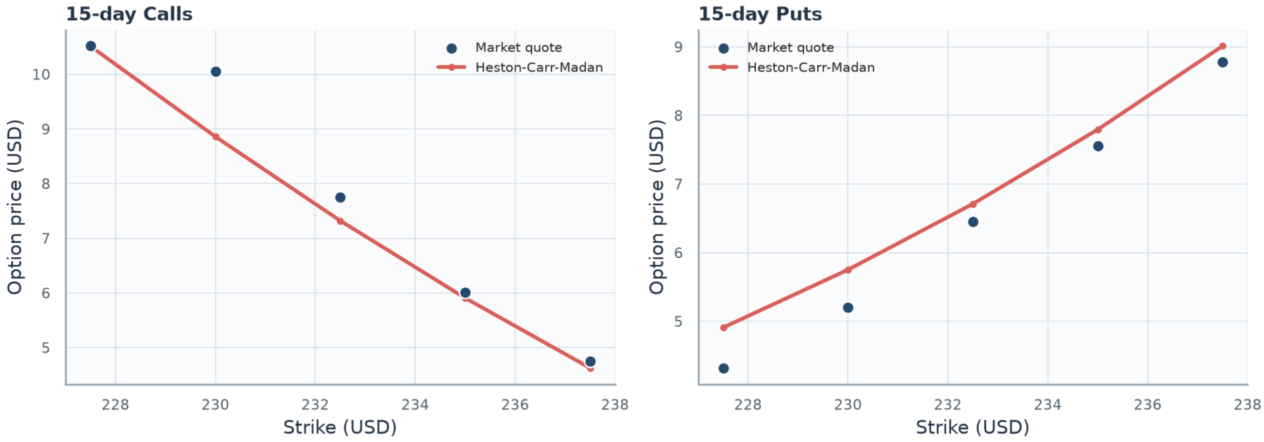
Parameters (Equations 8-9). psi_T(v) is the damped transform at Fourier frequency v and maturity T; alpha>0 is the damping parameter, r is the risk-free rate, and i=sqrt(-1). phi_log S_T is the characteristic function of the terminal log price. C_0(K,T) is today's call value, K is strike, and Re[dot] takes the real part of the integrand.

$$\psi_T(v) = \frac{e^{-rT}\phi_{\log S_T}(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}. \tag{8}$$

$$C_0(K,T) = \frac{e^{-\alpha\log K}}{\pi}\int_0^\infty \mathrm{Re}\left[e^{-iv\log K}\psi_T(v)\right]dv. \tag{9}$$

The damped transform follows Carr and Madan (61-73). Its MSE is 0.2458, compared with 0.2458 for Lewis. Fitted prices and numerical stability are the meaningful comparison.

Kappa changes from 2.2032 under Lewis to 0.0504 under Carr-Madan even though the losses are identical to four decimals. This is consistent with weak short-maturity identification: the two numerical formulas agree on prices, while the optimizer can move along a flat parameter direction.



Plot 3. 15-day Carr-Madan fit. The market points and fitted curves nearly reproduce the Lewis result, showing agreement in prices despite materially different calibrated parameters.

1(iii) 20-day ATM Asian call

The arithmetic-average Asian payoff reduces dependence on a single terminal stock price. Monte Carlo simulation is required because the payoff depends on the complete daily path.

Parameters (Equations 10-13). A_T is the arithmetic average through maturity T; n=20 is the number of future daily observations, j indexes observation dates t_j, and S_t_j is the stock price at each date, including S_t_0=S_0. Pi_T is the Asian-call payoff and K=S_0 is its at-the-money strike. widehatV_0 is the discounted Monte Carlo value, M=200,000 is the number of paths, m indexes paths, Pi_T^(m) is path m's payoff, and r is the risk-free rate. V_client is the quoted value after multiplying fair value by 1.04 to include the 4% fee.

$$A_T = \frac{1}{n+1}\sum_{j=0}^{n} S_{t_j}, \qquad S_{t_0} = S_0. \tag{10}$$

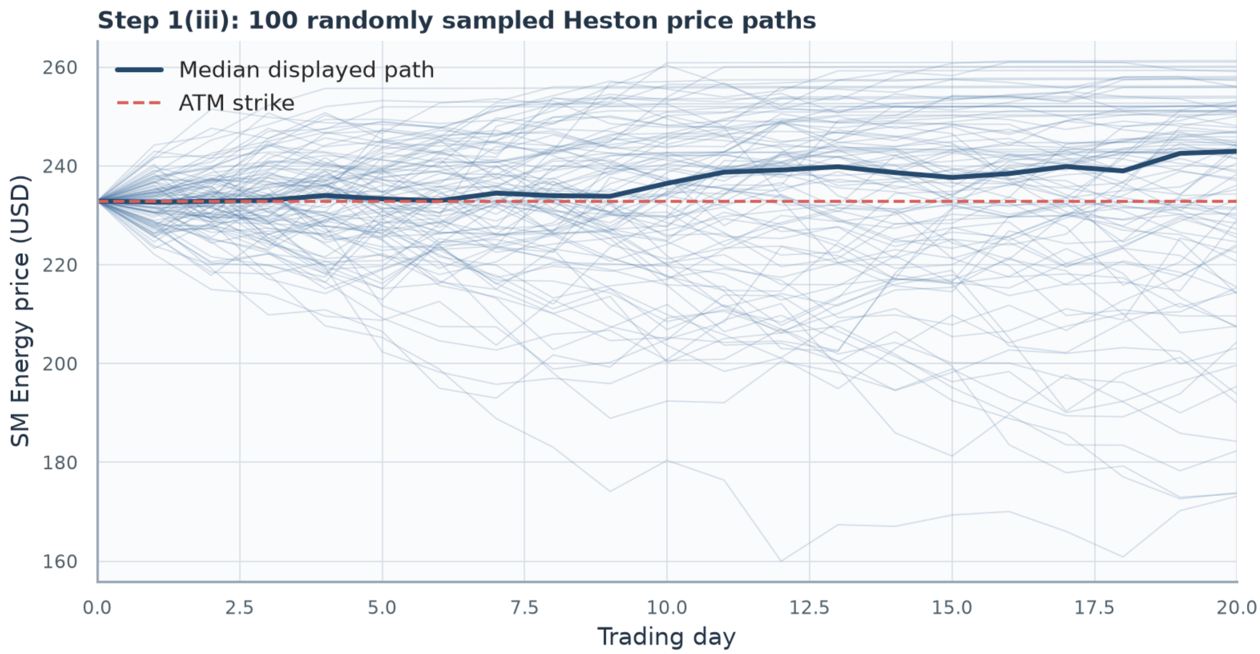
$$\Pi_T = \max(A_T - K, 0), \qquad K = S_0. \tag{11}$$

$$\widehat{V}_0 = e^{-rT}\frac{1}{M}\sum_{m=1}^{M} \Pi_T^{(m)}. \tag{12}$$

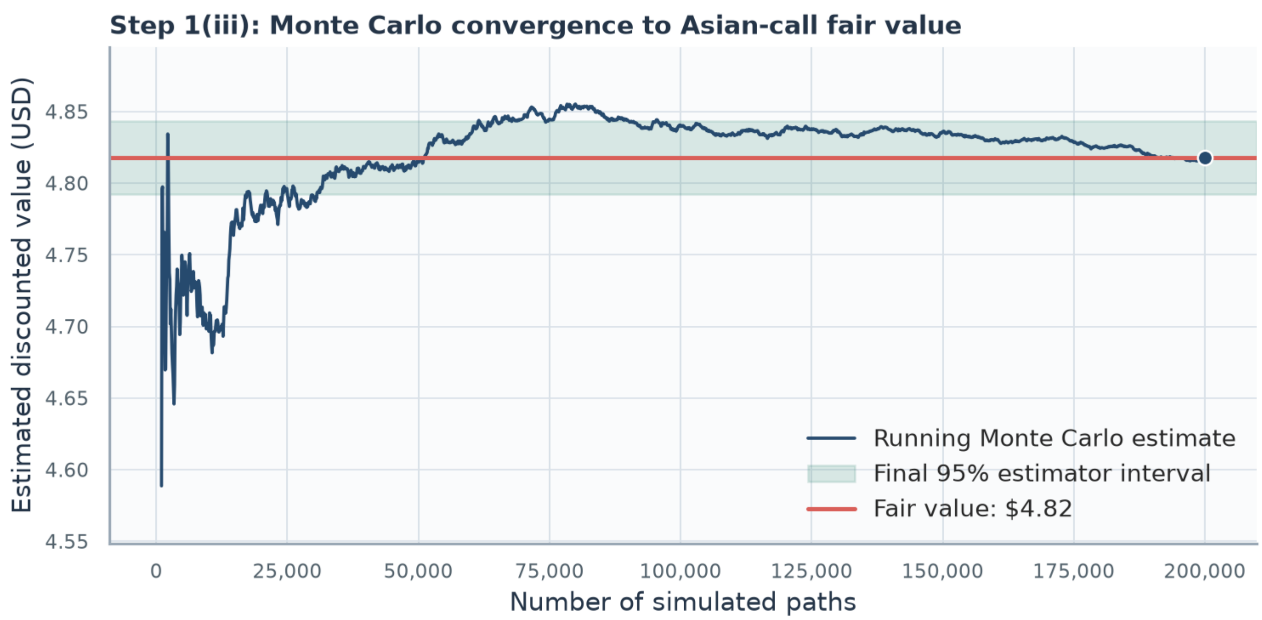
$$V_{\text{client}} = 1.04\,\widehat{V}_0. \tag{13}$$

Today's spot is included in the arithmetic average, following the project instructions (WorldQuant University 1-2). Across 200,000 risk-neutral scenarios, fair value is $4.8177 (standard error $0.0130), with a 95% estimator interval of $4.7922 to $4.8432. The 4% fee produces a client price of $5.0104.

The standard error is 0.27% of fair value, so simulation noise is small relative to calibration risk. The interval measures estimator precision and is not a guaranteed payoff range for the client.



Plot 4. 100 randomly sampled Asian-call paths. The paths illustrate the range of simulated stock trajectories, the highlighted median path gives a typical scenario, and the strike line marks the payoff threshold.



Plot 5. Asian-call fair-value convergence. The running estimate settles near the final fair value as the number of paths increases, while the confidence band narrows as Monte Carlo uncertainty declines.

Step 2 - 60-day derivative

2(i) Bates calibration using Lewis

Bates extends Heston with random discontinuous price jumps, allowing the model to represent rare abrupt moves in addition to ordinary stochastic-volatility dynamics.

Parameters (Equations 14-16). S_t^- is the stock price immediately before a jump; r, v_t, and W_t^S retain their Heston meanings. N_t is a Poisson jump count with intensity lambda, J=e^Y is the multiplicative jump size, and Y is normal with mean mu_J and standard deviation delta_J. The compensator kappa_J=E[J-1] preserves the risk-neutral drift. phi_J(u) is the jump characteristic-function multiplier at Fourier argument u over maturity T, with i=sqrt(-1).

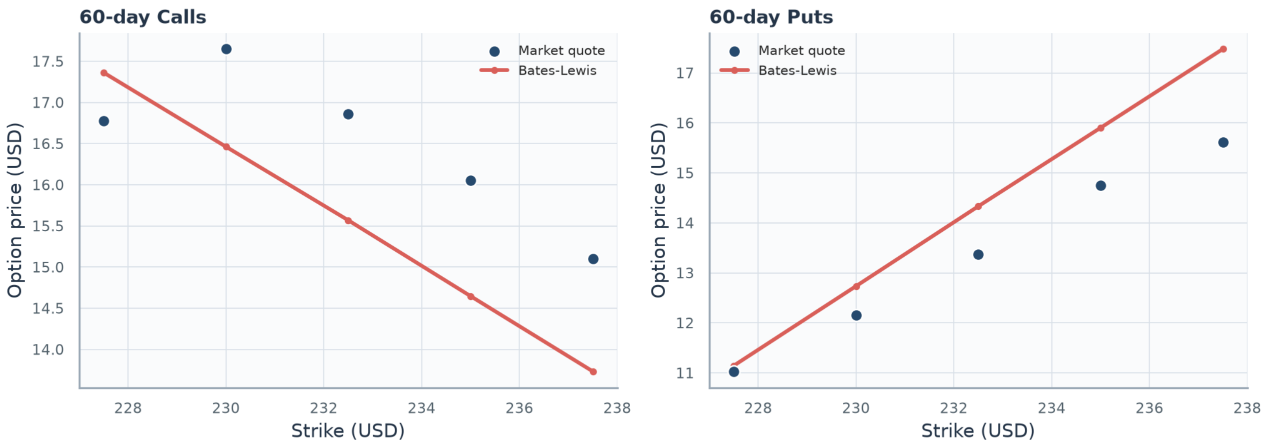
$$\frac{dS_t}{S_{t^-}} = (r-\lambda\kappa_J)\,dt + \sqrt{v_t}\,dW_t^S + (J-1)\,dN_t. \tag{14}$$

$$\kappa_J = \mathbb{E}[J-1] = \exp\left(\mu_J + \frac{\delta_J^2}{2}\right) - 1. \tag{15}$$

$$\varphi_J(u) = \exp\left\{\lambda T\left[\exp\left(iu\mu_J - \tfrac{1}{2}\delta_J^2 u^2\right) - 1 - iu\kappa_J\right]\right\}. \tag{16}$$

The 60-day jump-diffusion calibration follows Bates (69-107) and has MSE 1.3324 and RMSE $1.1543.

The call-put pairs have parity RMSE $2.2533. Rho is at its upper bound, theta and initial variance are at lower bounds, and jump volatility is at its lower bound. These boundary estimates and a Feller diagnostic of -0.0228 show that the eight parameters are weakly identified by this small, internally inconsistent cross-section.



Plot 6. 60-day market and model prices. The Bates curves capture the overall strike pattern, but visible gaps between market points and model values show the limitations of fitting the inconsistent call-put cross-section.

2(ii) Carr-Madan comparison

This specification retains the Bates jump and variance dynamics but uses the Carr-Madan transform as a numerical cross-check of Lewis pricing.

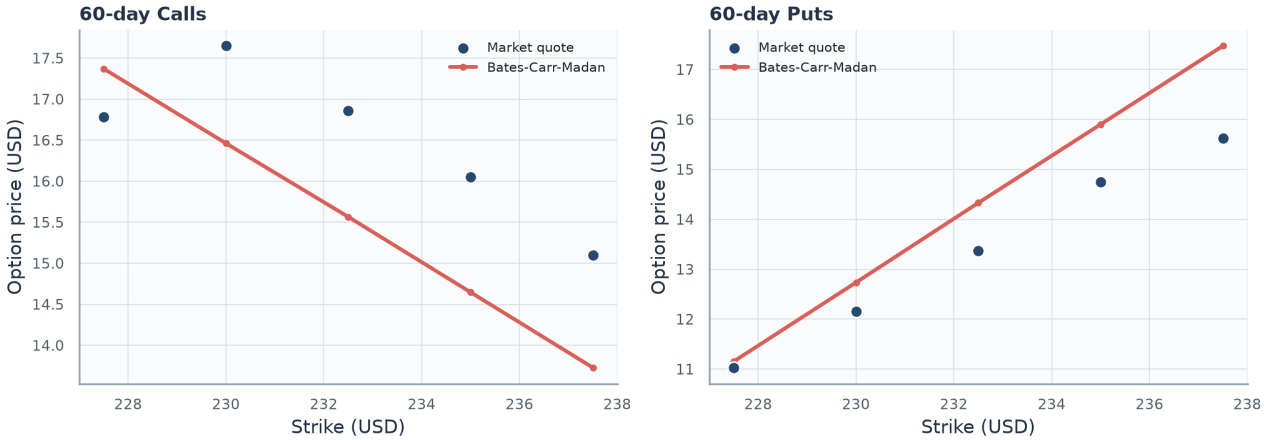
Parameters (Equations 8-9). psi_T(v) is the damped transform at Fourier frequency v and maturity T; alpha>0 is the damping parameter, r is the risk-free rate, and i=sqrt(-1). phi_log S_T is the characteristic function of the terminal log price. C_0(K,T) is today's call value, K is strike, and Re[dot] takes the real part of the integrand.

$$\psi_T(v) = \frac{e^{-rT}\,\phi_{\log S_T}(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}. \tag{8}$$

$$C_0(K,T) = \frac{e^{-\alpha\log K}}{\pi}\int_0^\infty \mathrm{Re}\left[e^{-iv\log K}\psi_T(v)\right]dv. \tag{9}$$

The matching Carr-Madan calibration follows Carr and Madan (61-73) and has MSE 1.3335. The comparison is based on fitted-price errors and numerical stability.

Jump intensity and mean jump size remain stable across methods, but kappa changes from 0.0114 to 14.9006. Nearly equal loss with very different mean reversion indicates a flat or multi-modal objective; fitted prices are more reliable than parameter equality.



Plot 7. 60-day Carr-Madan fit. The transform-based fit reproduces the broad market-price shape and closely matches the Lewis pricing result, supporting numerical consistency across methods.

2(iii) 70-day 95%-moneyness put

The project specifies a strike of 95% of spot (WorldQuant University 3), giving $221.25. Fair value is $8.8187 and the price after the 4% fee is $9.1715.

Parameters (Equation 17). K is the put strike, S_0=232.90 is the current SM Energy spot price, and 0.95 is the assignment's 95% moneyness factor, giving K=221.255.

$$K = 0.95S_0 = 0.95(232.90) = 221.255. \tag{17}$$

The 70-day maturity is a ten-trading-day extrapolation beyond the calibration sample. The extension is modest, but the boundary-sensitive Bates estimates make calibration risk more material than the difference between the two Fourier integration methods.

Step 3 - Euribor rates

3(i) CIR calibration

CIR models a non-negative, mean-reverting short rate and connects its parameters to the Euribor curve through closed-form zero-coupon bond prices.

Parameters (Equations 18-23). r_t is the short rate at time t, r_0 is its initial value, a is mean-reversion speed, b is the long-run rate, eta is rate volatility, and W_t is a Brownian motion. P(0,T) is today's zero-coupon bond price for maturity T; A(T) and B(T) are the CIR affine bond coefficients and gamma=sqrta^2+2eta^2 is an auxiliary coefficient. R(0,T) is the continuously compounded zero rate, while exp and log denote the exponential and natural logarithm.

$$dr_t = a(b - r_t)\,dt + \eta\sqrt{r_t}\,dW_t. \tag{18}$$

$$P(0,T) = A(T)\exp[-B(T)r_0]. \tag{19}$$

$$B(T) = \frac{2(e^{\gamma T}-1)}{(\gamma + a)(e^{\gamma T}-1)+2\gamma}. \tag{20}$$

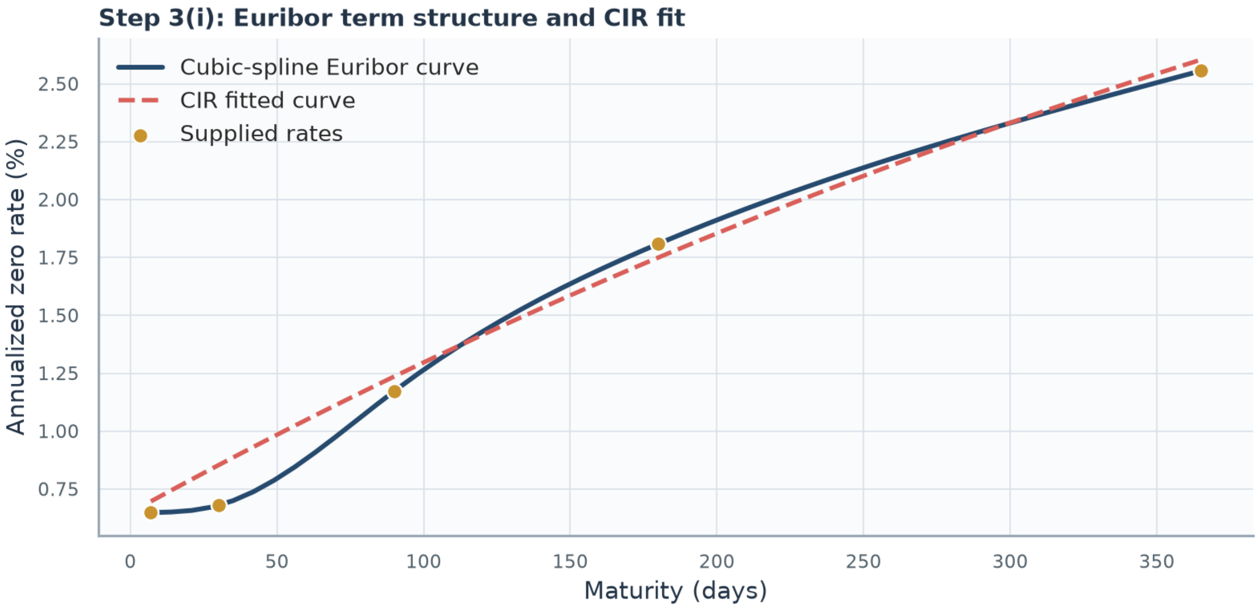
$$A(T) = \left[\frac{2\gamma\exp((a+\gamma)T/2)}{(\gamma + a)(e^{\gamma T}-1)+2\gamma}\right]^{2ab/\eta^2}. \tag{21}$$

$$\gamma = \sqrt{a^2 + 2\eta^2}. \tag{22}$$

$$R(0,T) = -\frac{\log P(0,T)}{T}. \tag{23}$$

The supplied curve contains 0.648% (1 week), 0.679% (1 month), 1.173% (3 months), 1.809% (6 months), and 2.556% (12 months). Weekly interpolation follows the project brief (WorldQuant University 3-4), and the rate model follows Cox, Ingersoll, and Ross (385-407). The fitted values are a=0.7525, b=0.0742, eta=0.5638; MSE is 0.00000062.

Mean reversion implies a half-life of 0.92 years, while the long-run rate 7.4170% lies above current 12-month Euribor. Curve RMSE is 7.90 basis points, but the Feller diagnostic -0.2063 is negative, so the process can reach zero and the broad dynamics remain uncertain.



Plot 8. Euribor curve and CIR fit. Observed tenor points rise with maturity, and the CIR curve follows that upward term-structure shape while leaving small deviations summarized by the curve RMSE.

3(ii) One-year rate scenarios

Daily CIR simulation turns the calibrated rate dynamics into a one-year distribution, supporting an expected-rate estimate and an explicit measure of rate uncertainty.

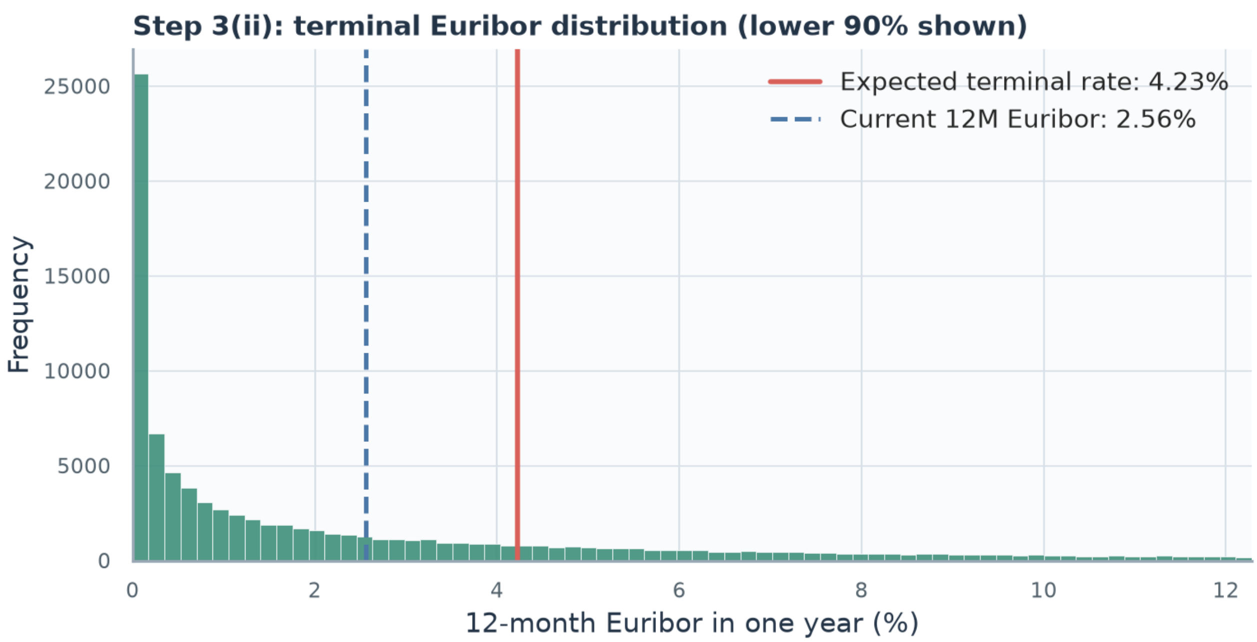
Parameters (Equations 24-25). r_1y is the simulated terminal rate after one year and Q_p(dot) is the p-quantile operator, so Q_0.025 and Q_0.975 form the central 95% range. estimated E[r_1y] is the Monte Carlo estimate of the expected terminal rate; M=100,000 is the number of paths, m indexes paths, and r_1y^(m) is path m's terminal rate.

$$[Q_{0.025}(r_{1y}),\ Q_{0.975}(r_{1y})]. \tag{24}$$

$$\widehat{\mathbb{E}}[r_{1y}] = \frac{1}{M}\sum_{m=1}^{M} r_{1y}^{(m)}, \qquad M = 100{,}000. \tag{25}$$

Following the required simulation size (WorldQuant University 4), 100,000 daily scenarios give an expected terminal 12-month Euribor of 4.2289%; the 95% range is 0.0000% to 25.0856%. Higher expected rates increase discounting of future positive cash flows and generally lower present values, all else equal.

The expected rate is 167.3 basis points above today's 12-month quote. The distribution is strongly right-skewed: the lower endpoint is pinned at zero while the upper tail reaches high rates. Product effects are not universal because rates influence both discounting and risk-neutral drift; the CIR short rate is used here as a proxy for 12-month Euribor rather than a full multi-curve tenor model.



Plot 9. Zoomed terminal Euribor distribution. The histogram concentrates near low rates and has a long right tail; the current-rate and expected-rate lines show the simulated upward shift, while the zoom excludes only the highest 10% from view.

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